

Archit Vishnoi

Portfolio: <https://v-archit.github.io> | Location: Rochester, NY

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Skills

5+ Years Unity Engine, C#, Data Structures, Github, Visual Studio, Logic, Scripting, Windows.

3+ Years in Unreal Engine, C++, Jira, 3D Maths, Android publishing, Performance Optimization.

1+ Years OpenGL, DirectX, HLSL, Perforce, Trello, Engine Programming, Agile Development, iOS Publishing,

<1 Years MySQL, ASP.NET Core, MAUI, Blazor Hybrid, Kestrel, Blazor WebAssembly, REST api

Experience

Newagez Technologies LLC, Missouri

Jan 2025 – Present

Software Engineer

- Designed interactive gameplay systems in Unity using C#, including physics-based movement, collision detection, and rigidbody manipulation for responsive character control and ragdoll simulation.
- Built UI components such as health bar, cooldown indicator, mini-map, scores, ammo count, and crosshair.
- Developed low-latency networking modules in native C++ for multiplayer synchronization over UDP and integrated with Unity via C# sockets, achieving sub-50 ms roundtrip latency in a simulated WAN environment.
- Created custom tools to streamline level design and assets management, improving team productivity.

Vizara Technologies Pvt Ltd, Gurugram, India

May 2019 – Jun 2021

Computer Graphics Engineer

- Spearheaded the end-to-end product development lifecycle for innovative applications, including *Digital Gandhi Akar Prakar* and *ChariotVR*, achieving successful launches on the Google Play Store and Apple App Store.
- Boosted the Museum AR app's stability by 20% through memory optimization and resolving unreferenced co-routines, leveraging Unity Profiler to diagnose and address performance bottlenecks.
- Acted as Scrum Master, driving Agile adoption by leading sprint planning, code reviews, facilitating team collaboration, maintaining Git codebase and ensuring seamless communication using Jira.
- Optimized GPU-intensive rendering processes for STL and FBX models in Unity3D, improving performance.
- Achieved timely goals with a 100% completion rate and translated user stories into actionable sprint tasks.
- Represented Vizara at international events, including Unity Developer Conference 2020, showcasing products.

Indian Institute of Technology, Delhi, India

Aug 2018 – Apr 2019

Research Fellow

- Engineered 3D visualization techniques in Unity3D, resulting in a 50% improvement in user immersiveness.
- Led the development of 'Digital Mini Spectacle', an interactive software project funded by the Ministry of Science.

Projects

Into The Babyverse – 3D Space Shooter Game – Unreal C++

M.S. 4th Sem

- Created a 6-degree-of-freedom spaceship system, custom hit physics, and a Chaos destruction framework
- Designed and implemented a Dynamic Spline Management system for object picking, achieving a 25% reduction in rendering time, and improving performance efficiency.
- Utilized Agile methodology in processes, including maintaining a comprehensive Game Design Document and conducting iterative evaluations through Sprints, ensuring consistent project alignment and timely delivery.

Hoodwink (Group) – 3D Wild West Action Game – Unreal C++

M.S. 3rd Sem

- Designed and implemented dynamic AI systems for Townsfolk NPCs, including patrolling, chasing, melee combat, and fleeing behaviors, improving gameplay realism.
- Developed and optimized Behavior Trees (BT) with custom BT Tasks, Decorators, and multi-sensory Perception Graphs (audio, visual, and touch) to enhance NPC decision-making and interactivity.

The AI Story (Group) – 2D Narrative Game – Unity C# Inky

M.S. 2nd Sem

- Led development of programming, UI, logic creation, and dialogue systems with iterative sentence display.
- Streamlined team communication, boosting collaboration and reducing project timelines by 15%.

Tower Defense – 2D Realtime Strategy Game – Unity C#

M.S. 1st Sem

- Enhanced AI to prioritize battles dynamically, increasing resource utilization and optimizing capacity deployment.
- Engineered a battle system ensuring 100% accuracy in combat status tracking and reducing processing latency.

Education

M.S. Game Design Development

Aug 2021 – Dec 2023

Rochester Institute of Technology, Rochester, NY

- Key coursework: Game Development Process, Engine and Graphics Programming, and AI.

B.Tech. Computer Science Engineering

Aug 2014 – May 2018

Jaypee Institute of Information Technology, Noida, India